

Generation Interconnection System Impact Study Report

For Queue Project Q21-1877

Honeyville 138 kV

March 04, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1877. The Interconnection Customer proposes a 181.9 MW solar photovoltaic facility interconnecting at the Honeyville 138 kV substation in the North Carolina area. The request was received on September 03, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 566 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$215,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Brimley – Mecklenburg Power Sta 138 kV	Tecumseh Energy Center 345-138 kV TX	108.5	348	61.4

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Bimley – Mecklenburg Power Sta 138 kV	Reconductor line	\$1,083,000	20% (\$215,000)
	Total allocated to project		\$215,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.